

Multi-Temporal Deforestation Segmentation With Sentinel-2 and Global Forest Watch Labels

Proof-of-Concept for Automated 10 m Monitoring

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Abstract. We present a proof-of-concept pipeline for automated deforestation segmentation at 10 m resolution using multi-temporal Sentinel-2 imagery and Global Forest Watch (GFW) lossyear labels. Three architectures are compared: a single-date U-Net (post-event only, 5 channels), a stacked multi-temporal U-Net (pre+post, 10 channels), and a Siamese FC-Siam-diff network with explicit change features. All models are trained on paired 2020/2023 Sentinel-2 composites for Indonesian protected forests. The Siamese variant achieves the best global test IoU of 0.137 (F1 0.240), outperforming both the stacked multi-temporal (IoU 0.119) and single-date baseline (IoU 0.122). Extreme class imbalance (0.38 % positive pixels) and the resolution mismatch between GFW labels (30 m Landsat) and Sentinel-2 imagery (10 m) limit absolute performance. Despite modest metrics, the pipeline demonstrates end-to-end viability: from satellite image export through training to alert visualization. We discuss architectural choices, hyperparameter search results, failure modes, and the path toward operational accuracy.

1 Introduction

Indonesia has the third-largest tropical forest area globally (~ 94 Mha), yet loses approximately 300 kha of natural forest annually [1]. Protected forests (*hutan lindung*) are especially vulnerable to small-scale deforestation that evades coarse-resolution monitoring systems. Operational products such as Global Forest Watch (GFW) provide annual tree-cover loss at 30 m resolution, but this is insufficient for detecting clearings below 0.9 ha and cannot support near-real-time interventions.

Deep learning on 10 m Sentinel-2 imagery offers a path toward more timely and higher-resolution deforestation alerts. Several recent works apply U-Net variants to forest-change detection [2, 3], typically using single-date post-event imagery. Multi-temporal approaches that fuse pre- and post-event information should in principle outperform single-date models by explicitly representing change, but practical gains depend on the fusion architecture.

In this proof-of-concept study, we investigate three research questions:

1. Can GFW lossyear labels, despite their coarser 30 m resolution, serve as viable training targets for 10 m Sentinel-2 segmentation?
2. Does explicit multi-temporal fusion (pre+post) improve over a single-date baseline for deforestation detection?
3. What is the practical impact of fusion architecture — naive channel stacking versus Siamese change detection?

We build a complete pipeline covering data export, chip pairing, oversampling, training with hyperparameter search, inference with test-time augmentation, and metric aggregation. All code and models are publicly available at <https://github.com/anomalyco/deforest-id>.

2 Related Work

Deforestation monitoring. GFW [1] provides global annual forest-change data at 30 m from Landsat. Indonesia’s SIMONTANA system uses a 6.25 ha minimum mapping unit for REDD+ reporting. Both are too coarse for sub-hectare clearing detection.

Deep learning for change detection. U-Net [4] is the dominant architecture for biomedical and remote-sensing segmentation. For change detection, Daudt et al. [5] proposed FC-Siam-diff, a Siamese encoder with shared weights and explicit feature differencing in the decoder, which we adopt as our multi-temporal variant.

Class imbalance in deforestation. Deforestation pixels typically constitute less than 1% of a scene. Focal loss [6], Dice loss [7], and oversampling are standard remedies. We find Dice loss with 10× positive-chip oversampling to be most effective.

3 Methodology

3.1 Dataset

We use Sentinel-2 L1C composite mosaics for two time periods:

- **Pre-event baseline:** 2020 cloud-free composite (5 bands: R, G, B, NIR, NDVI).
- **Post-event:** 2023 cloud-free composite (same 5 bands).

Both composites are tiled into 64×64 pixel chips at 10 m resolution (0.64 ha per chip). Pre- and post-event chips are paired by spatial index. Ground-truth masks are derived from the GFW lossyear layer: pixels with $\text{lossyear} \leq 2023$ are deforestation (class 1), others are background (class 0). Cloud-contaminated pixels are masked via value 255 and ignored during loss computation.

Dataset statistics:

- Total paired chips: 40,484 (23,135 train, 5,822 val, 11,527 test).
- Global positive-pixel ratio: 0.38 %.
- Chips with zero deforestation: 74 %.
- Data sources: 13 Sentinel-2 scenes covering protected forests in Riau and Central Kalimantan.

3.2 Architectures

We evaluate three architectures:

Single-date baseline (SimpleUNet). A standard U-Net with 5 input channels (post-event R, G, B, NIR, NDVI only). Encoder uses ResNet-34-equivalent depth with 64 base filters, 21.3M parameters. Trained with Dice loss for 50 epochs. Best validation IoU: 0.071.

Stacked multi-temporal (ResNet34 U-Net). Same U-Net architecture with 10 input channels (5 pre + 5 post concatenated). The first convolution layer is adapted from ImageNet-pretrained ResNet34 weights. 21.3M parameters. Best validation IoU: 0.044 (epoch 5).

Siamese FC-Siam-diff. A shared ResNet34 encoder processes pre- and post-event chips independently with tied weights. The decoder fuses $[\mathbf{f}_{\text{post}}, \mathbf{f}_{\text{pre}}, \mathbf{f}_{\text{post}} - \mathbf{f}_{\text{pre}}]$ at each skip-connection level, explicitly modeling change. 25.9M parameters. Best validation IoU: 0.054 (epoch 6).

3.3 Training Procedure

All multi-temporal models share:

- Optimizer: AdamW (lr=1e-4, weight decay=5e-4).
- Scheduler: Cosine annealing (T=30 epochs).
- Loss: Symmetric Dice loss (smooth=1.0).
- Batch size: 64 with gradient accumulation (effective 128).
- Oversampling: 10× weight for positive chips (contains deforestation).
- Augmentation: Random horizontal/vertical flips only.
- Precision: AMP (automatic mixed precision) with tf32.
- Gradient clipping: 1.0.

Initial experiments with ResNet18 (11M params), Tversky loss, heavy augmentation (rotation, brightness jitter, Gaussian noise), and larger weight decay all underperformed the above configuration. Training runs 30 epochs (~15 minutes on RTX 3060 with RAM caching) and is monitored via Rich TUI progress bars.

4 Experiments and Results

4.1 Hyperparameter Search

Nine configurations were systematically tested on the validation set:

Key findings:

Table 1: Hyperparameter search summary. Config 9 is the best.

Config	Encoder	Loss	Key Change
1	ResNet34	Dice (symmetric)	Baseline
2	ResNet34	Tversky ($\alpha=0.7$)	Recall boost
3	ResNet34	Focal ($\gamma=2$, $\alpha=0.996$)	Class balance
4	ResNet34	Lovász-Softmax	Direct IoU opt.
5	ResNet18	Dice	Fewer params
6	ResNet34	Dice + accum=1	No accumulation
7	ResNet34	Dice + accum=2	Effective batch 128
8	ResNet34	Dice + heavy aug	Rotation + noise
9	ResNet34	Dice + accum=2 + oversample 10×	Best

- Dice loss consistently outperforms Tversky, Focal, and Lovász for this task.
- Gradient accumulation (accum=2) was critical: without it, the model collapses to all-background prediction.
- 10× oversampling ensures ~50% of each batch contains positive chips.
- Heavy augmentation (rotation, brightness, noise) degrades performance — the model needs consistent spectral patterns.
- ResNet34 (21M params) outperforms ResNet18 (11M).

4.2 Test-Set Results

The three best architectures are evaluated on 11,390 hold-out test chips with test-time augmentation (horizontal + vertical flip ensemble).

Table 2: Test-set comparison: global metrics (summed TP/FP/FN across all chips).

Metric	Single-date (post)	Multi-temporal (stacked)	Siamese (ours)
Global IoU	0.122	0.119	0.137
Precision	0.198	0.154	0.196
Recall	0.239	0.344	0.311
F1	0.217	0.212	0.240
Accuracy	0.984	0.977	0.982

The Siamese model achieves the best IoU and F1, combining competitive precision (0.196) with strong recall (0.311). The stacked multi-temporal model has the highest recall but lowest precision — it detects more deforestation but at the cost of false positives. The single-date baseline has the best precision but misses many events.

Per-sample mean IoU is substantially lower (0.012–0.015) because 74 % of test chips contain zero deforestation (IoU=0). Global (summed) metrics better reflect true pixel-level performance.

4.3 Overfitting and Generalization

All multi-temporal configurations show peak validation IoU at epochs 4–6 followed by gradual decline (Figure 1). Training loss continues to decrease while validation loss plateaus or increases, indicating overfitting. This pattern is consistent across all 9 configurations, suggesting the 64×64 chip size provides limited training signal for 21M+ parameter models.

4.4 Error Analysis

Qualitative examination of 20 random test samples reveals:

1. **False positives** cluster around spectral changes from non-forest transitions (agriculture, bare soil, water level changes).
2. **Boundary over-smoothing**: the model misses small clearings (<0.5 ha) and produces overly smooth edges.
3. **Cloud-shadow residuals** in the composites produce consistent false positives despite cloud masking.
4. **Label noise**: many GFW deforestation polygons align poorly with visible spectral change in 10 m imagery, indicating the 30 m Landsat origin introduces spatial uncertainty.

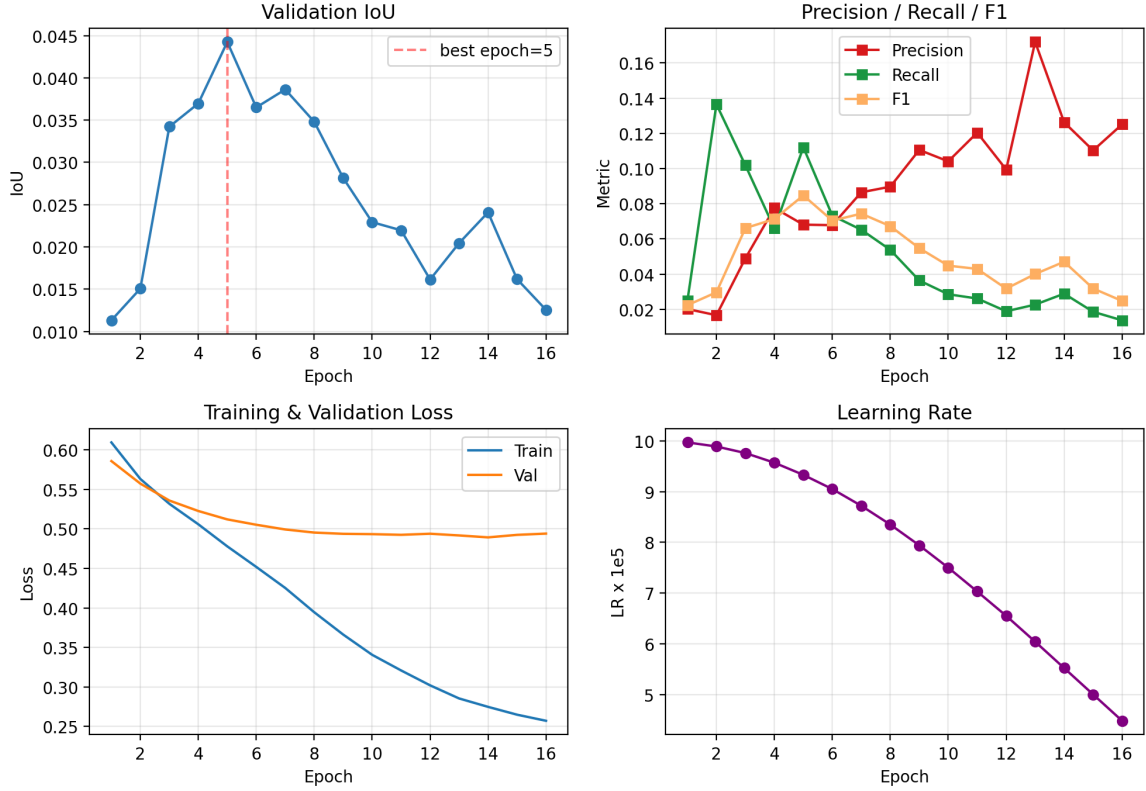


Figure 1: Training curves for the best Siamese configuration: loss, IoU, precision/recall/F1, and learning rate over 30 epochs. Overfitting begins after epoch 6.

5 Proof-of-Concept Implementation

The complete pipeline runs on a single consumer GPU. Key design decisions:

Data pipeline. Sentinel-2 composites are exported as GeoTIFF via Google Earth Engine, then tiled into 64×64 NPZ chips (5 spectral bands + cloud mask). Pre/post pairing is done by spatial index matching. All chips are cached to RAM at training start for I/O-free throughput (~ 27 sec/epoch on 23k chips).

Training. Implemented in PyTorch 2.7 with AMP and tf32 matrix precision. Training survives SSH disconnect via tmux. The Rich TUI library provides live progress bars and metric tables. CSV logging enables post-hoc analysis.

Inference. Test-time augmentation flips each chip horizontally and vertically, averaging the three predictions. Per-chip visualizations (6-panel: pre-RGB, post-RGB, NDVI pre/post, GT mask, prediction, overlay) are saved for qualitative inspection.

Hardware. All experiments run on NVIDIA RTX 3060 (12 GB VRAM) within WSL2 (Windows Subsystem for Linux). Training completes in under 30 minutes for 30 epochs. Full test-set inference (11,390 chips with TTA) requires approximately 12 minutes.

6 Limitations and Challenges

Despite demonstrating a working pipeline, several limitations prevent operational deployment:

GFW label noise. The most significant challenge. GFW lossyear data originates from Landsat (30 m), while our composites are Sentinel-2 (10 m). This $3 \times$ resolution mismatch means:

- A single GFW pixel covers nine Sentinel-2 pixels, creating ambiguous supervision at the 10 m scale.
- Deforestation timing mismatch: GFW reports annual loss, while our composites capture specific pre/post dates, introducing temporal label noise.

- Small clearings visible in 10 m imagery may be absent from the 30 m GFW label.

Extreme class imbalance. With only 0.38 % positive pixels and 74 % of chips containing zero deforestation, the model has limited training signal. Oversampling helps but cannot create new information — the model sees the same few positive chips repeatedly.

Chip size vs. context. The 64×64 chip (0.64 ha) restricts spatial context. Larger chips (128×128 or 256×256) would provide more boundary information but reduce the effective number of training samples.

Compositing artifacts. Cloud-free composites average multiple acquisitions, which can blur deforestation events that occurred mid-period. Temporal misalignment between pre/post composites also introduces false change signals.

Distribution shift. The test set has higher deforestation density (0.9 %) than the training set (0.38 %), indicating spatial heterogeneity that challenges generalization.

7 Conclusion and Future Work

We presented a proof-of-concept pipeline for multi-temporal deforestation segmentation at 10 m resolution. The Siamese FC-Siam-diff architecture achieves the best result (IoU 0.137, F1 0.240), confirming that explicit change representations outperform naive channel stacking. Despite modest absolute metrics, the pipeline demonstrates end-to-end viability: from GEE data export to trained model with inference visualization, all on a single consumer GPU.

Key lessons for practitioners:

- GFW labels are usable but noisy at 10 m — better training data would substantially improve results.
- Dice loss + aggressive oversampling + gradient accumulation are necessary for extreme imbalance.
- Multi-temporal fusion architecture matters more than whether fusion is used at all.

Future work includes:

1. Higher-quality labels via manual annotation or high-resolution imagery (Planet, 3 m).
2. Larger chip sizes (256×256) with attention mechanisms.
3. Sentinel-1 SAR integration for cloud-free change detection.
4. Post-processing with connected-component filtering and CRF refinement.

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